What demographic, financial, and relationship factors are associated with divorce?



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Introduction

Although every marriage is unique, large datasets can reveal common patterns associated with whether relationships remain intact or end in divorce.

This project examines demographic, financial, lifestyle, and relationship characteristics across 45,000 marriage records.

Our Dataset

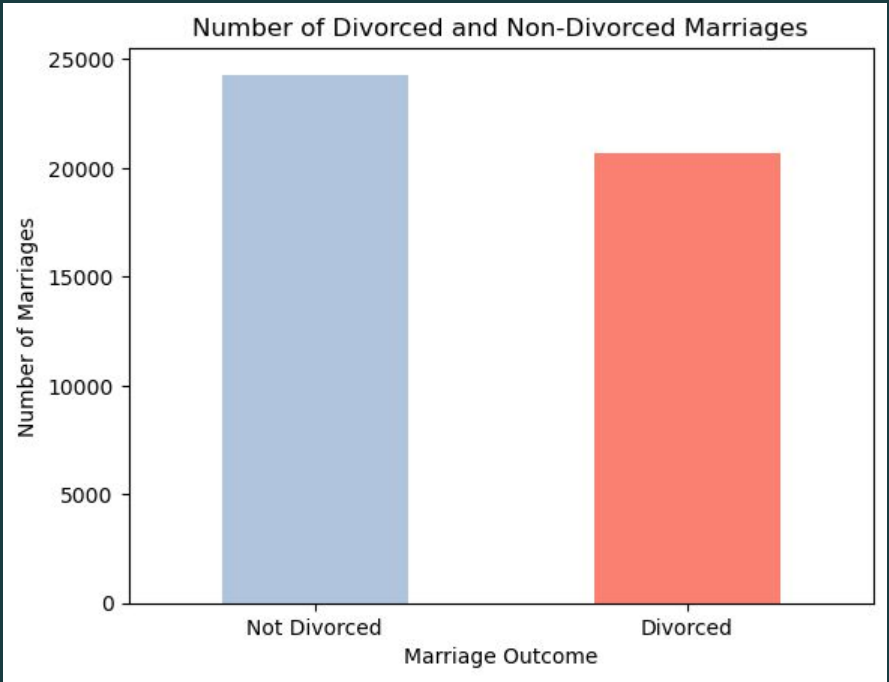
The Kaggle Marriage Longevity: What Makes Relationships Last dataset contains **45,000 marriage records** and **24 variables,** including age, education, income, financial stress, children, communication behaviors, shared activities, and divorce status.

Research Question

What demographic, financial, and relationship factors are associated with divorce?

Why It Matters

Divorce can affect emotional well-being, household finances, family stability, and children. Identifying patterns may provide useful insights for couples, counselors, researchers, and educators.

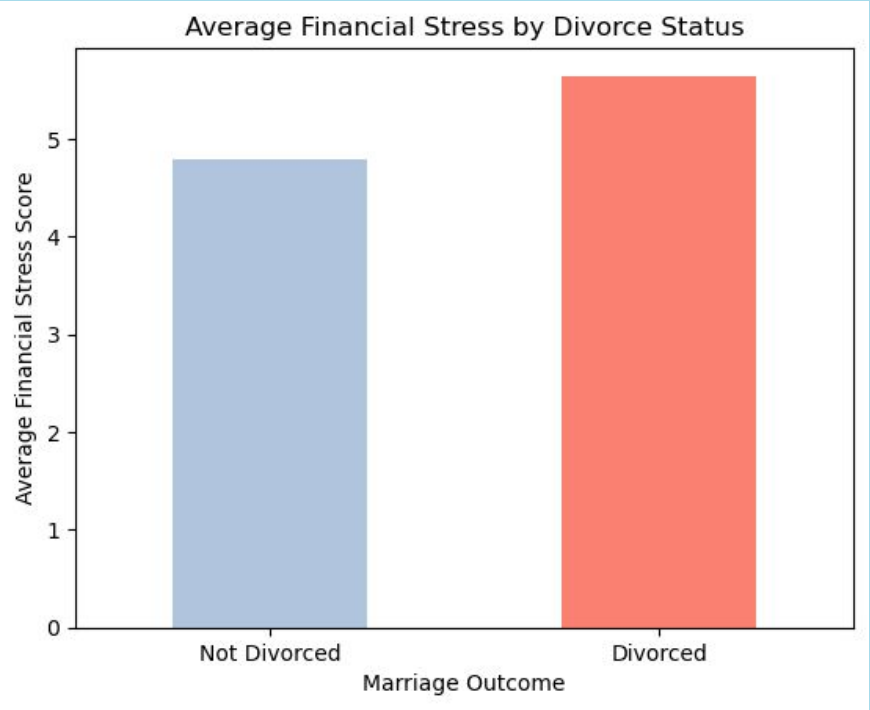


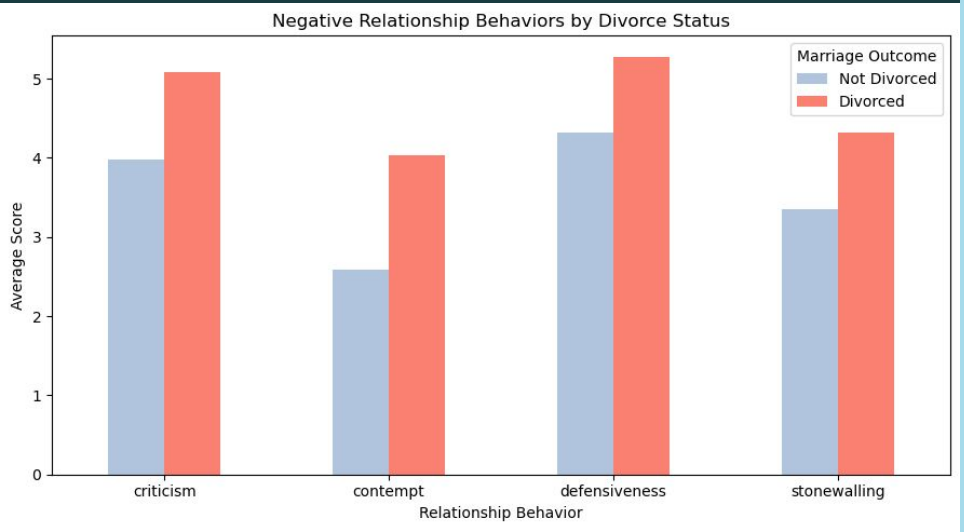
Overall Divorce Rate

- Approximately 46% of marriages ended in divorce
- Dataset contains a balanced mix of divorced and non-divorced couples
- Establishes a reliable baseline for comparing the two groups

Financial Stress

- Divorced couples reported higher financial stress
- Financial strain is associated with divorce
- Financial factors appear important but are not the strongest predictors



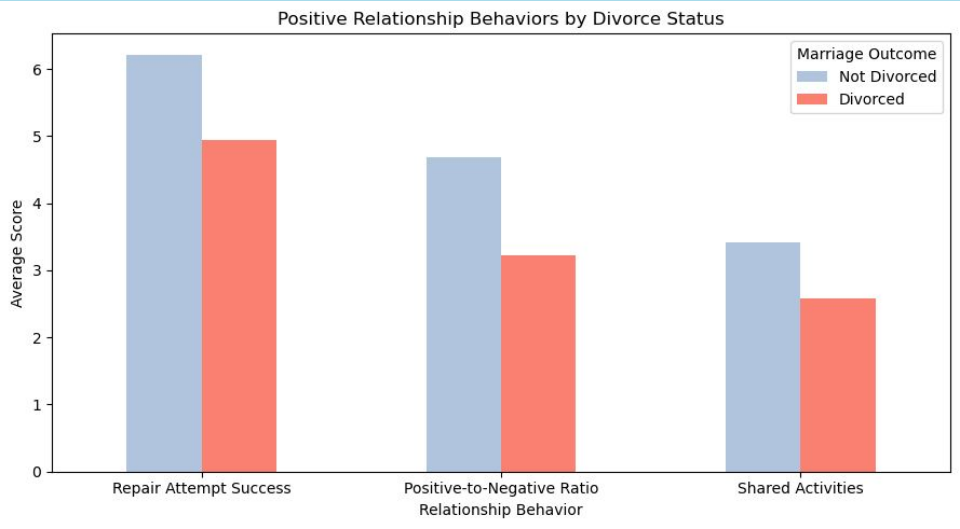


Negative Relationship Behaviors

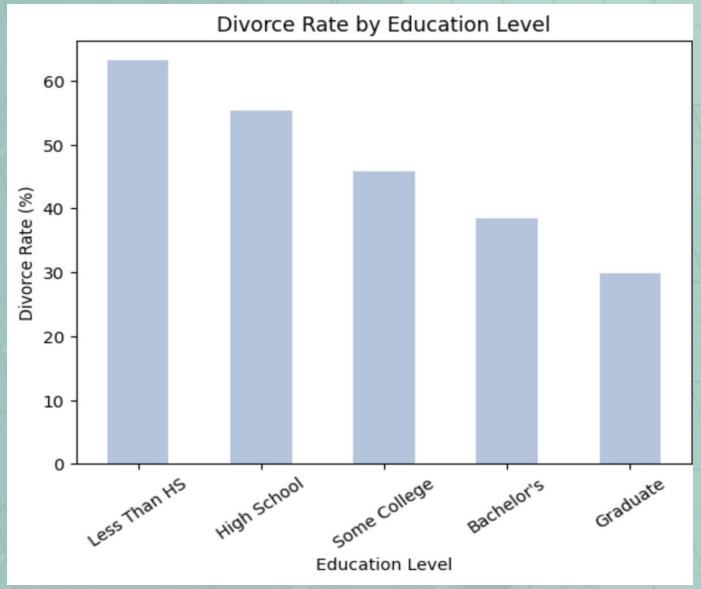
- Divorced couples reported more criticism, contempt, defensiveness, and stonewalling
- Contempt showed the biggest difference between the two groups
- Defensiveness had the highest average score overall
- Overall, negative communication patterns were more common among divorced couples

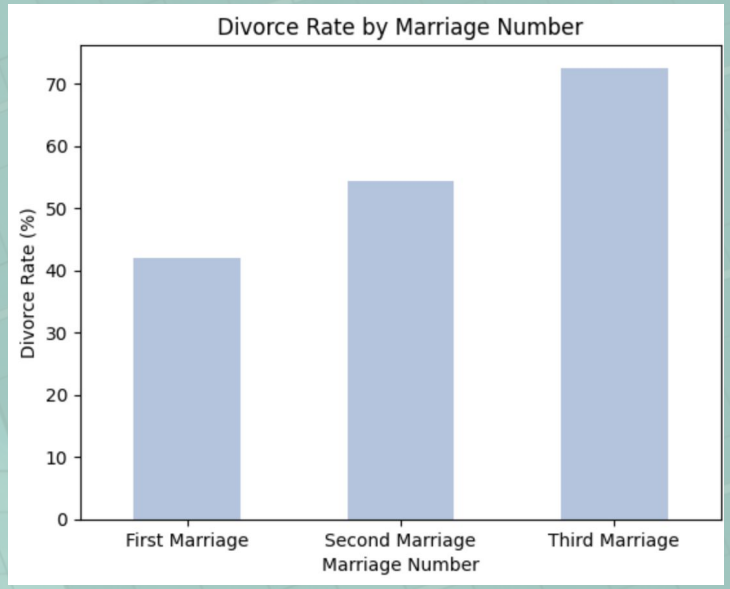
Positive Relationship Behaviors

- Non-divorced couples had more successful repair attempts
- They also had a higher positive-to-negative interaction ratio
- Non-divorced couples reported more shared activities each week
- Overall, positive communication and shared time were more common among couples who remained married

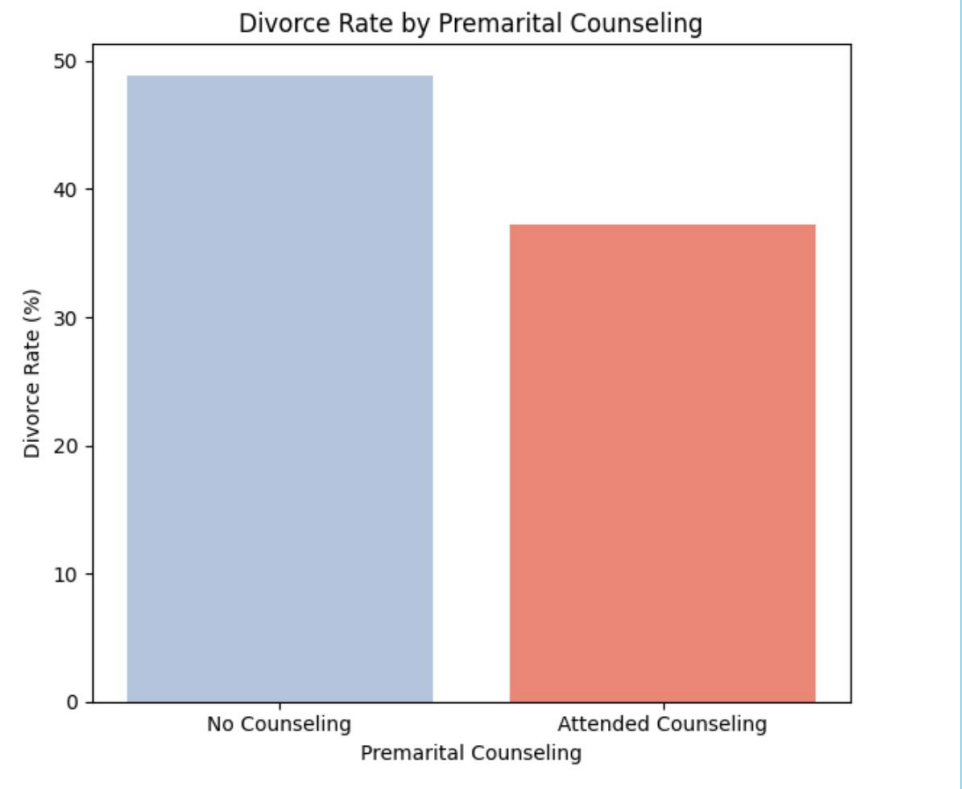


Education & Marriage Number





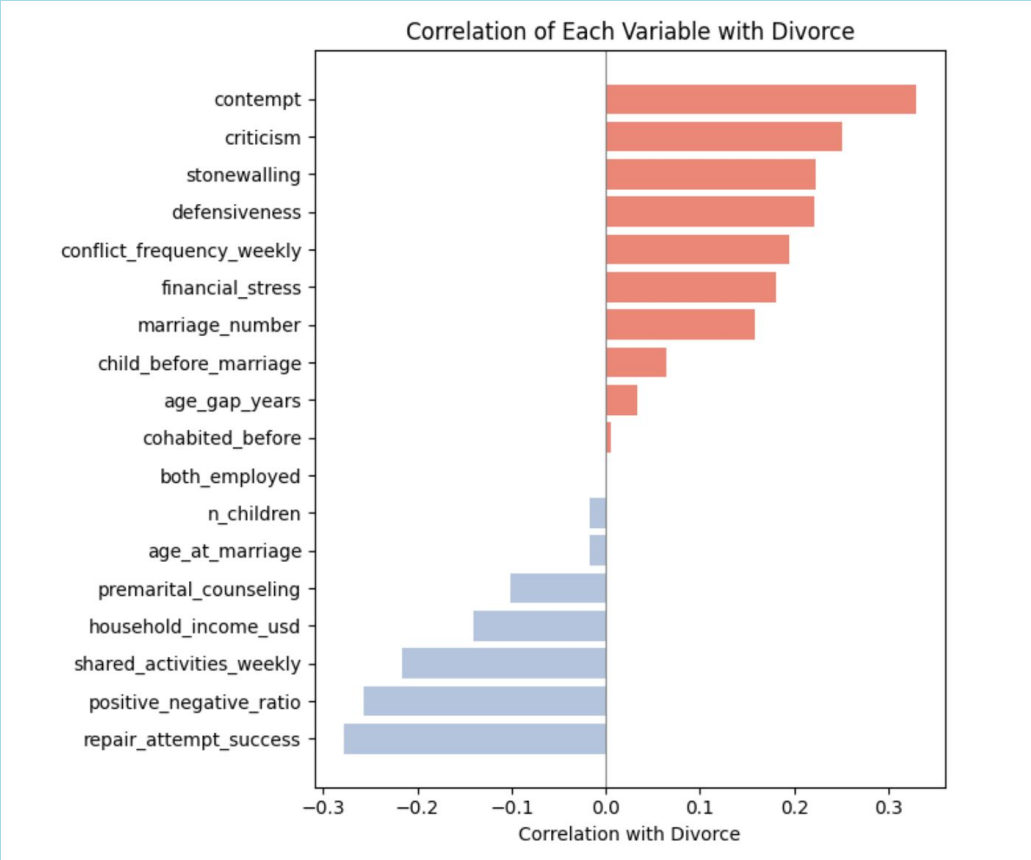
- Divorce rates decreased as education increased
- Later marriages had progressively higher divorce rates
- Demographic factors matter, but relationship behaviors showed stronger associations overall



Premarital Counseling

- Couples attending counseling had an 11.7 percentage-point lower divorce rate
- Even just attempting counseling lowered divorce rates
- **Unlike most variables, this is something couples can actively choose**

Correlations



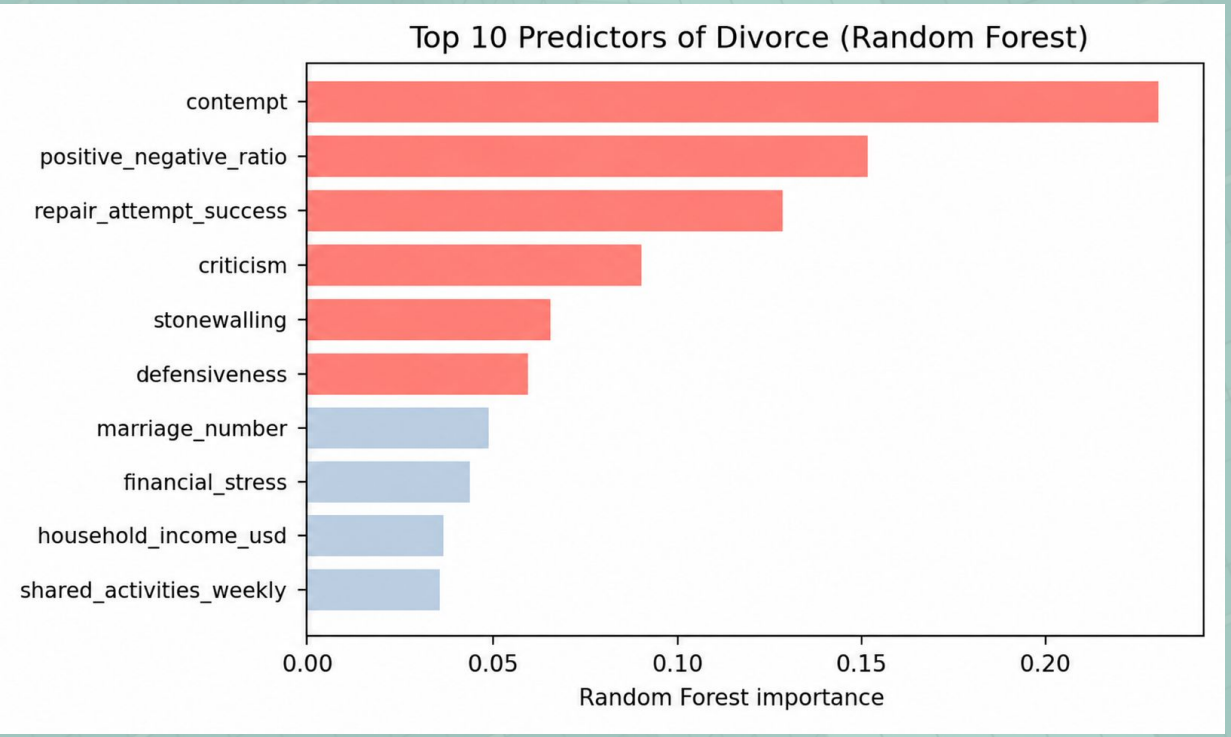
- Contempt had the strongest positive correlation (r = 0.33)
- Repair-attempt success and positive interactions had the strongest negative correlations
- Cohabitation, employment, and children showed little association

Classification Analysis

Model	Accuracy	AUC
Logistic Regression	**75.7%**	**.837**
Random Forest	**73.5%**	**.812**

- Both models outperformed the 54% divorce rate baseline
- Logistic Regression performed slightly better
- The results support the patterns observed in the exploratory analysis

Random Forest Feature Importance

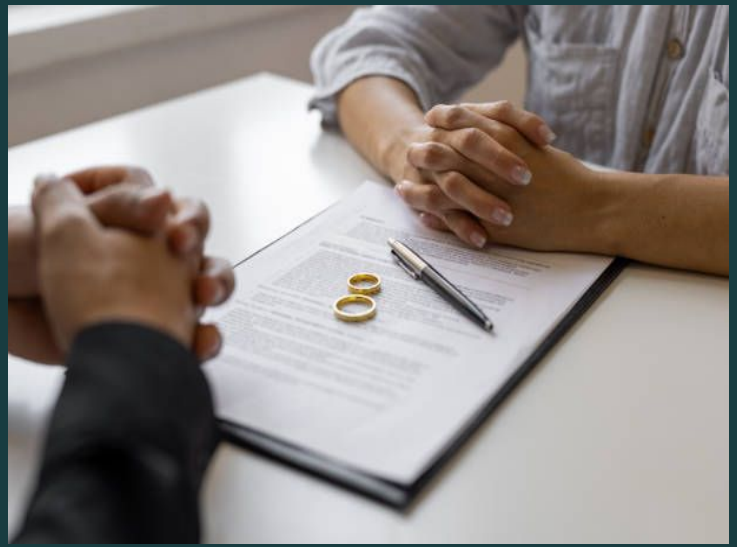


- Contempt ranked as the most important predictor
- Positive-to-negative ratio and repair-attempt success followed closely
- Relationship behaviors dominated demographic variables

Conclusions

Relationship behaviors showed the strongest associations with divorce, more so than most demographic or financial factors.

- Communication mattered most: Contempt stood out as the strongest predictor, while successful repair attempts and more weekly positive interactions were associated with staying married.
- Other factors still mattered: Financial stress and later marriages were associated with higher divorce rates; education and premarital counseling were associated with lower rates.
- The models supported our exploratory analysis: Logistic Regression reached 75.7% accuracy, and Random Forest again ranked contempt as the most important predictor.



Thank you!
Questions?